

❖ Diseases of Mulberry Silkworm

1. Viral Disease

➤ Symptom:

- ✓ The larvae will be sluggish with swollen intersegmental region.
- ✓ The integument of diseased larvae will be fragile and breaks easily.
- ✓ On injury milky fluid containing many polyhedral inclusion bodies oozes out from the larval body.
- ✓ The diseased larvae do not settle for moult and show shining integument.
- ✓ The larvae appear to be restless.
- ✓ The dead larvae hang by hind legs heads downward

➤ Management

- ✓ Sun drying of rearing appliances for one/two days.
- ✓ Disinfection of rearing room and appliances with 5% bleaching powder.
- ✓ Disinfection of worms, trays and discarding of diseased worms
- ✓ Ensure proper ventilation and air circulation.
- ✓ Provide proper bed spacing.
- ✓ Feed the larvae with nutritious mulberry leaves.
- ✓ Collect and burn infected larvae, faecal matter and bed refuses.
- ✓ Early diagnosis and rejection of infected lots.
- ✓ Dust the bed disinfectant, Vijetha (or) Resham Keet Oushadh on the larvae, after each moult and ½ hr. before resumption of feeding (3 kg/100 dfl).
- ✓ Spray 1% of extract of *Psoralea coryleifolia* on mulberry leaves, shade dry and feed worms once during third instars.

2. Bacterial Diseases

- Bacteria and viruses cause the disease individually or in combination.
- Fluctuating temperature and humidity and poor-quality mulberry predispose the disease development.
- The diseased larvae will be stunted in growth, will be lethargic soft and appear flaccid.
- The cephalothoracic region may be translucent.
- The larvae vomit gut juice develop dysentery and excrete chain type feces.

- The larvae on death putrefy, develop different and emit foul smell

- **Management**

- ✓ Maintenance of hygienic condition.
- ✓ Disinfection of rearing room and appliance.
- ✓ Disinfection of worms, trace and discarding of sick wormsAvoid injury to the worms, overcrowding of trays and accumulation of faeces in the rearing bed.
- ✓ Sound management, improving the rearing environment and feed stuff.
- ✓ Feeding the larvae with healthy nutritious leaves.
- ✓ Early diagnosis and rejection of infected lots.
- ✓ Avoid spraying commercial **Bt** insecticides in nearby mulberry field.
- ✓ Apply antibiotics like Streptomycin/Tetracycline/Ampicillin

i. White muscadine

- It is caused by a fungus *Beauveria bassiana* and the green muscadine is caused by a fungus *Spicaria prasina*.
- Aspergillosis is common in young age silkworms and the infected larvae will be lustrous and die.
- Dark green (*Aspergillus flavus*) or rusty brown (*Aspergillus tamari*) mycelial cluster are seen on the dead body.
- The diseases larvae prior to death will be lethargic and on death are flaccid oil specks may be seen on the surface of larvae.
- They gradually be foam hard, dry, and mummify into a white or green colored structure.
- The diseases pupae will be hard, lighter and mummifies.
- **Management:**
 - ✓ Sundry the rearing appliances.
 - ✓ Disinfect the rearing room and utensils with 5 per cent bleaching powder.
 - ✓ Avoid low temperature and high humidity in the rearing room.
 - ✓ Keep the rearing bed thin and dry.
 - ✓ Early diagnosis and rejection of infected lots.
 - ✓ Apply Dithane M45 (3 kg/100 dfls) / Vijetha supplement as disinfectant on the larvae.
 - ✓ Disinfect rearing rooms and trays with 4 per cent pentachlorophenol to control Aspergillosis.

3. Protozoan disease:

i. Pebrine

- Diseases larvae show slow growth, undersized body and poor appetite.
- Diseases larvae reveal pale and flaccid body. Tiny black spots appear on larval integument.
- Dead larvae remain rubbery and do not undergo putrefaction shortly after death.
- **Management of Pebrine:**
 - ✓ Produce healthy eggs.
 - ✓ Disinfection of rearing room and utensils.
 - ✓ Maintain strict hygienic conditions during rearing
 - ✓ Surface disinfect the laying in 2 per cent formalin for 10 minutes before incubation.
 - ✓ Collect and burn the diseased eggs, larvae, pupae and moths, bed refuses, faecal pellets, etc.